

# AMEN ALLAH YOUSSEF

AI / ML Research Engineer — LLM Systems, Retrieval-Augmented Generation, Applied Model Evaluation

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## PROFILE

ML engineer with 1.5+ years of experience across internship, freelance, and applied engineering roles, specialized in LLM systems and retrieval-augmented generation, with a research-first approach: every architectural choice (retrieval strategy, model size, fine-tuning method) is justified against an evaluation protocol, not intuition. Built RAG pipelines, decoder-only transformers, and RL fine-tuning (LoRA, PPO, GRPO) from first principles, with systematic ablations tracked in Weights & Biases. Pursuing a Master's in Intelligent Pervasive Systems focused on efficient fine-tuning and retrieval-augmented generation.

## TECHNICAL SKILLS

**Languages:** Python, NumPy, SQL, TypeScript

**ML / DL:** Transformer architecture (decoder-only), causal self-attention, PPO, GRPO, SFT, LoRA fine-tuning, RLHF, BPE tokenization, reward shaping, curriculum learning

**LLM & RAG:** RAG pipeline design, hybrid SQL + vector retrieval, vector search (Qdrant, ChromaDB, AstraDB), reranking, semantic caching, agentic orchestration, multi-hop query decomposition

**Evaluation & Experimentation:** Recall@k, MRR, NDCG, Faithfulness, Precision@k, bootstrap CI, Weights & Biases (run tracking, hyperparameter sweeps), evaluation-first / leakage-free protocol design

**Frameworks:** PyTorch, JAX/Flax, Hugging Face Transformers, scikit-learn, SentenceTransformers, FastAPI, Docker

## EXPERIENCE

**AI Engineer** · Novera

- Architected an AI-powered photo-matching system (InsightFace face recognition) deployed on Hugging Face Spaces, integrated into a production PostgreSQL pipeline.
- Contributed to open-source AI projects within a 3-person engineering collective, covering LLM API integration, agentic workflow design, and inference optimization.

**AI Engineering Intern** · Mobelite

- Designed and shipped a production RAG system (AstraDB + Gemini API) for semantic search across 500+ company documents, cutting average lookup time by ~65% vs. a timed manual baseline.
- Benchmarked 4 chunking strategies and 3 embedding configurations against a precision@5 evaluation set; the selected configuration improved retrieval precision by ~30% over the initial deployment.

**Freelance AI Engineer — Voice Automation** · Independent

- Designed and deployed an LLM-powered voice agent handling 10,000+ outbound calls/month at 99.5% uptime, scaling client outreach ~10x without added headcount.

## RESEARCH & APPLIED PROJECTS

**Grade Advisor — Hybrid SQL + Vector RAG for Steel Grade Selection** | [GitHub](#)

Python · DuckDB · ChromaDB · Gemini API · Pydantic

- Built on Erasteel's own ASP® technical datasheets: designed a hybrid retrieval architecture combining deterministic SQL filtering (exact composition and hardness constraints) with vector search (qualitative intent), after establishing that pure vector RAG is unsafe for numeric engineering constraints.
- Built a multimodal extraction pipeline turning 13 raw PDF datasheets into a validated structured schema, with automated physical-sanity checks and a deterministic offline fallback for zero-downtime operation.
- Designed a zero-tolerance benchmark (18 test cases: exact constraint match, precision@3, impossible-query rejection); reached 100% pass rate.

**ArXiv Research Assistant — Evaluation-First RAG over 5,000 Papers** | [GitHub](#)

Python · PyTorch · Qdrant · SentenceTransformers · FastAPI · Weights & Biases

- Built a full RAG pipeline (embed → search → rerank → generate) reaching 0.962 faithfulness and 0.992 citation accuracy on a held-out 25-query test set.
- Ran a systematic embedding and reranking ablation (bge-small vs. e5-small vs. MiniLM) tracked in W&B; froze the corpus and split train/val/test before labeling to prevent leakage.

**TacticsGPT — Decoder-Only Transformer for Football Tactical Analysis** | [GitHub](#)

Python · NumPy · PyTorch · JAX · LoRA · GRPO · Weights & Biases

- Implemented a decoder-only transformer from scratch (causal self-attention, positional encoding, autoregressive decoding) without high-level abstractions.
- Pretrained on a custom BPE tokenizer, then applied LoRA SFT followed by GRPO-style RL fine-tuning, improving task reward by 41% over the SFT baseline.

**Turbine Energy Yield Prediction — Applied ML Case Study** | [Project](#)

Python · scikit-learn · Pandas

- Framed a gas-turbine energy-yield regression problem from real sensor data: engineered leakage-free features and validated on a time-based split to avoid look-ahead bias.
- Trained and tuned baseline and random-forest models, reported MAE/RMSE/R<sup>2</sup> honestly against held-out data, and authored a one-page model card covering limitations and failure modes.

## EDUCATION

**Master's in Intelligent Pervasive Systems** · ISSATSO, Sousse — Agentic AI, RL, Efficient Fine-Tuning

**Bachelor's in Software Engineering** · ISSATSO, Sousse

## LANGUAGES

Arabic (Native) · English (C1) · French (Intermediate–Good, B1–B2)